



# Increase of vitamin D<sub>2</sub> by UV-B exposure during the growth phase of white button mushroom (*Agaricus bisporus*)

HanneL. Kristensen, Eva Rosenqvist & Jette Jakobsen

To cite this article: HanneL. Kristensen, Eva Rosenqvist & Jette Jakobsen (2012) Increase of vitamin D<sub>2</sub> by UV-B exposure during the growth phase of white button mushroom (*Agaricus bisporus*), Food & Nutrition Research, 56:1, 7114, DOI: [10.3402/fnr.v56i0.7114](https://doi.org/10.3402/fnr.v56i0.7114)

To link to this article: <https://doi.org/10.3402/fnr.v56i0.7114>



© 2012 Hanne L. Kristensen



Published online: 02 Apr 2012.



Submit your article to this journal [↗](#)



Article views: 1082



View related articles [↗](#)



Citing articles: 2 View citing articles [↗](#)

# Increase of vitamin D<sub>2</sub> by UV-B exposure during the growth phase of white button mushroom (*Agaricus bisporus*)

Hanne L. Kristensen<sup>1\*</sup>, Eva Rosenqvist<sup>2</sup> and Jette Jakobsen<sup>3</sup>

<sup>1</sup>Department of Food Science, Aarhus University, Aarslev, Denmark; <sup>2</sup>Department of Agriculture and Ecology, Copenhagen University, Copenhagen, Denmark; <sup>3</sup>National Food Institute, Technical University of Denmark, Copenhagen, Denmark

## Abstract

**Background:** Mushrooms are the only non-animal food source of vitamin D. Wild mushrooms have naturally high vitamin D<sub>2</sub> content, and cultivated mushrooms produce vitamin D<sub>2</sub> from ergosterol when exposed to supplementary UV-B during the post-harvest phase.

**Objectives:** This study investigated the effects of providing supplementary UV-B during the growth phase on vitamin D<sub>2</sub> formation and the interactions with growth of mushrooms, as compared to supplementary UV-B during the post-harvest phase or exposure to sunlight for both cultivated and wild mushrooms.

**Methods:** Experiments were carried out with exposure to supplementary UV-B just prior to harvest in the range of 0–2,400 mJ cm<sup>-2</sup>. Mushrooms grew for 2 days with or without repeated UV-B exposure each day. Vitamin D<sub>2</sub> and growth rate were determined. Some mushrooms were post-harvest treated by exposure at 200 mJ cm<sup>-2</sup> supplementary UV-B or natural sunlight, prior to vitamin D<sub>2</sub> determination.

**Results:** The content of vitamin D<sub>2</sub> was 0.2–164 µg 100 g<sup>-1</sup> fresh weight (FW), and there was a linear relationship between UV-dose up to 1,000 mJ cm<sup>-2</sup> and vitamin D<sub>2</sub> content. The fast growth rate of the mushrooms diluted the vitamin D<sub>2</sub> from 24 to 3 µg 100 g<sup>-1</sup> within 2 days of exposure at 200 mJ cm<sup>-2</sup>. Following repeated UV-B exposure, vitamin D<sub>2</sub> increased to 33 µg vitamin D<sub>2</sub> 100 g<sup>-1</sup>. Growth was unaffected by UV-B. Post-harvest exposure to supplementary UV-B resulted in a higher vitamin D<sub>2</sub> content of 32 µg 100 g<sup>-1</sup> compared to the 24 µg 100 g<sup>-1</sup> obtained from exposure to UV-B during the growth phase. In contrast, wild and cultivated mushrooms with and without exposure to sunlight had vitamin D<sub>2</sub> content in the range of 0.2–1.5 µg vitamin D<sub>2</sub> 100 g<sup>-1</sup>.

**Conclusions:** This study showed that mushrooms with a well-defined content of vitamin D<sub>2</sub> can be obtained by exposure to supplementary UV-B just prior to harvest.

Keywords: *ergocalciferol*; *growth phase*; *post-harvest*; *sunlight*; *wild mushrooms*; *yield*

During winter, food is the main source of vitamin D, and fish and other animal products are considered important sources of vitamin D to fulfil daily requirement. Wild mushrooms have been recognised as the only non-animal food source of vitamin D with reports of high vitamin D<sub>2</sub> contents of 3–59 µg 100 g<sup>-1</sup> fresh weight in, for example, *Cantharellus* sp. and *Boletus* sp. (1, 2). The content of vitamin D in wild relatives of the cultivated button mushroom, *Agaricus bisporus*, has not yet been reported.

Vitamin D<sub>2</sub> is formed during UV exposure, as a result of photochemical cleavage of the B ring of ergosterol forming the intermediate pre-vitamin D<sub>2</sub>, this then undergoes thermal re-arrangement to ergocalciferol (vitamin D<sub>2</sub>) (3).

Several recent studies have reported the same response in mushrooms exposed to supplementary UV post-harvest. They showed that exposure to UV-B (280–315 nm) was more effective in the formation of vitamin D<sub>2</sub> than was exposure to UV-A or UV-C (4, 5). A linear relationship between the UV dose and the vitamin D<sub>2</sub> formed was found, and vitamin D<sub>2</sub> in the mushrooms was found to be relatively stable during post-harvest storage (6). Based on this knowledge, commercial production of button mushrooms with a high content of vitamin D<sub>2</sub> was established based on post-harvest exposure to UV. Cultivated mushrooms are marketed as a fresh and natural product; therefore, to underpin this product profile, the UV treatment could be integrated into the growth phase.

Such treatment to naturally enhance vitamin D<sub>2</sub> in mushrooms prior to harvest would be more representative of the processes occurring in wild mushrooms exposed to sunlight. To relate supplementary UV exposure in mushrooms to natural conditions, more knowledge is needed on the interactions between mushroom growth, UV exposure, and vitamin D<sub>2</sub> as well as on the effect of natural sunlight on vitamin D<sub>2</sub> content in cultivated and wild mushrooms.

The objectives of this study were to investigate the effects of (1) one or repeated exposures of supplementary UV-B prior to harvest on vitamin D<sub>2</sub> formation as compared to post-harvest exposure, (2) mushroom growth rate on vitamin D<sub>2</sub> content and of UV exposure on growth rate and (3) exposure of cultivated mushrooms to sunlight as compared to wild mushrooms on vitamin D<sub>2</sub> content.

### Materials and methods

Two experiments were carried out with exposure of mushrooms to UV-B doses in the range of 0–250 mJ cm<sup>-2</sup> over three days (Experiment A), and exposure of mushrooms to UV-B doses of 220–2,400 mJ cm<sup>-2</sup> during one day (Experiment B).

#### The UV-B equipment

A transportable and adjustable UV-B unit was constructed using six UV-B tubes (Medical Therapy/12, Philips, Eindhoven, The Netherlands) giving a homogeneous distribution of UV-B over an area of 0.6 × 0.4 m<sup>2</sup>. The intensity of UV-B was 1.88 mW cm<sup>-2</sup> at a distance of 0.25 m between the mushrooms and the UV-B tubes. Exposure to UV-B for 1 min 46 sec provided a dose of 200 mJ cm<sup>-2</sup>. The spectral distribution of the UV-B unit was measured by a spectrometer (AvaSpec-2048, Avantes, Eerbeek, The Netherlands) that was calibrated with a standard of spectral irradiance OL FEL-C (Optronic Laboratories, Orlando, FL, USA), a lamp that includes wavelengths down to 250 nm. The spectral composition was measured at 0.25 m distance from the UV-B tubes, with a black screen surrounding the unit to avoid stray light. Measurements were an average of 10 scans (Fig. 1). The UV-B doses in the different treatments were measured by the accumulated intensity over time by use of a UV-B photometer (IL1400 SEL005/TLS312/Td; International Light Technologies, MA, USA).

#### Experimental design

White button mushrooms (*Agaricus bisporus* (Lange) Imbach) were grown at 18°C day and night temperature and 88% relative humidity on horse manure/barley straw compost at the commercial production facility of Egehøj Champignon A/S, Veflinge, Denmark in 2009. Experiment A was initiated on the second sampling day of the second flush (7-day harvest cycle), 36 days after

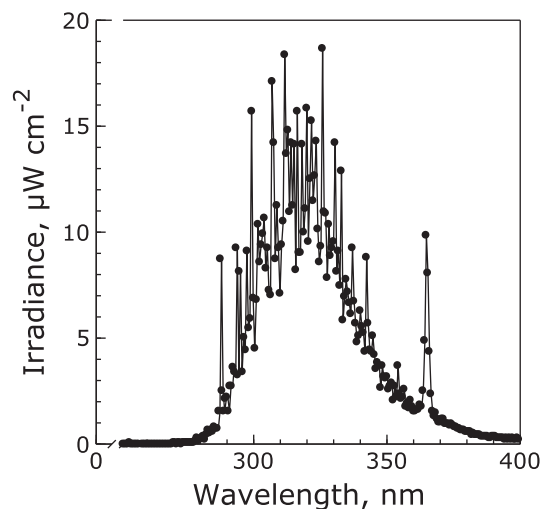


Fig. 1. The spectral distribution of the UV-B unit used for treatments of supplementary UV-B during the growth phase and post-harvest. The data are a mean of 10 scans and presented with 1.2 nm resolution.

inoculation of the compost with mycelium. Experiment B was carried out on the first day of the third flush, 45 days after inoculation. In both experiments, the mushrooms were sampled for yield and vitamin D<sub>2</sub> analyses at harvest, according to the sampling practice for commercial harvest-ready mushrooms.

In Experiment A, 28 plots of 0.4 × 0.4 m<sup>2</sup> were established on the shelves with growing mushrooms in a complete randomised statistical design with four blocks, two in each of two growth chambers giving four replicate plots per treatment. By varying the duration of exposure, the treatments were 0, 100, 150, 200 and 250 mJ m<sup>-2</sup> supplementary UV-B on the first day; and 200 mJ cm<sup>-2</sup> given repeatedly once a day on the first, second and third day of the experiment (3 × 200 mJ cm<sup>-2</sup>). All plots were sampled immediately after exposure and also on the three consecutive days (at 24 hour periods) after the first day exposure to 0 or 200 mJ cm<sup>-2</sup> and after the repeated exposure to 200 mJ cm<sup>-2</sup>. Additional samples were taken for post-harvest treatments of exposure to 200 mJ cm<sup>-2</sup> natural UV-B in sunlight for approximately 20 min at mid day, or following overnight cooling (2°C), re-adjustment to 18°C and exposure to 200 mJ cm<sup>-2</sup> supplementary UV-B. Additional production areas from the four blocks were sampled the three days for estimation of biomass of small mushrooms, that remained after harvest to grow for next day harvest. This was to calculate vitamin D<sub>2</sub> recovery during the experiment.

In Experiment B, seven plots of 0.6 × 0.4 m<sup>2</sup> were established on the shelves with growing mushrooms in a growth chamber. Treatments at 220, 500, 1,000 and 2,400 mJ m<sup>-2</sup> supplementary UV-B were given to the plots in a complete randomised statistical design with no replications in the range of 220–1,000 mJ m<sup>-2</sup> and six

replications of the 2,400 mJ m<sup>-2</sup> treatment. All plots were sampled immediately after exposure.

In the period of late July to October 2009, four composite samples of each 2–3 wild growing mushrooms (*Agaricus sp.*) were sampled in an orchard at Aarslev, Denmark (10°27'E, 55°18'N) for determination of vitamin D<sub>2</sub> content. In addition, four samples of *A. bisporus* of miscellaneous origin (strains and producers) were purchased for determination of vitamin D<sub>2</sub> content.

All fresh mushroom samples were weighed, cooled (1°C, 100% relative humidity) and mushrooms counted, cleaned and base removed. Each sample of at least 200 g material was frozen at -18°C for subsequent freeze drying for determination of dry matter and vitamin D<sub>2</sub> content.

#### Vitamin D<sub>2</sub> analysis

Prior to analysis, the frozen samples of mushrooms were thawed to room temperature, homogenised for 30 sec (Type 320, Moulinex, Group SEB, F) and freeze-dried (Christ Beta 1–8, Martin Christ Gefriertrocknungsanlagen GmbH, Osterode, D). To achieve a homogenised subsample, the freeze-dried mushrooms were homogenised again for 30 sec (Type 320, Moulinex, Group SEB, F).

The analytical method and the equipment used to determine vitamin D<sub>2</sub> in the mushrooms were a modification of a method used for meat (7). In short, 1 g freeze-dried mushrooms were taken for analysis, and vitamin D<sub>3</sub> was added as an internal standard according to EN12821. The samples were extracted by alkaline saponification followed by clean-up by silica solid phase extraction. Due to the extremely high amount of ergosterol in the samples, another preparative HPLC system was used than in meat (7). The combination of a silica and an amino column was replaced by a shorter silica column (Luna, Si 60, 3 µm, 150 × 4.6 mm, Phenomenex, Torrance, CA, USA) in combination with an isocratic mobile phase consisting of 2-propanol:methyl-tert-butyl-ether:cyclohexan:n-heptan (0.7:2:48.65:48.65). A flow rate of 1.2 ml min<sup>-1</sup> gave a retention time for vitamin D<sub>2</sub> of 11.2 min; sample collection was made during the interval 10.4–12.2 min. The retention time for ergosterol was 16.0 min, which was baseline separated from vitamin D<sub>2</sub>. For separation, a C18 column combined with a mobile phase consisting of acetonitrile:methanol (80:20) was used, whereas detection was performed by photodiode array detector (220–320 nm) and quantification at 265 nm. The analysis was performed as single determination. For quality control, a house-reference material consisting of freeze-dried mushrooms was analysed on each day of analyses, showing an internal reproducibility of 5.8%. The analysis was run in a laboratory accredited after ISO 17025 for the original analytical method, and the correctness for the method was controlled by

participation in the Food Analysis Performance Assessment Scheme ([www.fapas.com](http://www.fapas.com)).

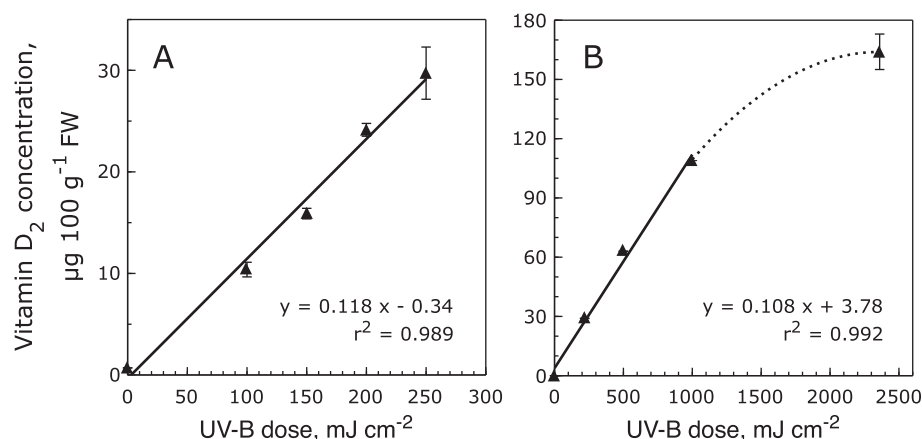
#### Data and statistical analysis

Recovery of vitamin D<sub>2</sub> produced after the first day exposure to UV-B was calculated per unit production area over the three days of experiment based on the biomass of large harvested and small remaining mushrooms. For this, it was assumed that vitamin D<sub>2</sub> content in small mushrooms equalled that was measured in harvested mushrooms.

The data were subjected to analysis of variance. Multiple comparisons were based on values of the least significant difference derived from analysis of variance (Proc GLM, SAS Institute Inc., Cary, NC, USA). When appropriate, the results were transformed before analysis to obtain homogeneity of variance. Differences determined at  $P < 0.05$  were considered statistically significant.

#### Results and discussion

Results from Experiments A and B showed a close to linear relationship between dose of UV-B and formation of vitamin D<sub>2</sub> at 0–1,000 mJ cm<sup>-2</sup> supplementary UV-B during the growth phase. The outputs from the linear regressions were very similar with a-factors of 0.118 and 0.108 for the range of 0–250 mJ cm<sup>-2</sup> and 0–1,000 mJ cm<sup>-2</sup>, respectively, and high correlation coefficients (Fig. 2A and B). The linearity of the relationship found in these two experiments was previously reported by Roberts et al. 2008 (6) following post-harvest exposure in the range of 500–1,000 mJ cm<sup>-2</sup> supplementary UV-B after overnight cooling of white button mushrooms. However, in the present study, the a-factor of the linear regression was 0.11 at exposure during the growth phase, compared to 0.06 calculated from Roberts et al. (2008) (6). Thus, exposure, for example, to 500 mJ cm<sup>-2</sup> during the growth phase formed vitamin D<sub>2</sub> of approximately 57 µg 100 g<sup>-1</sup> fresh weight compared to 37 µg 100 g<sup>-1</sup> at post-harvest exposure (6). However, the difference between the two studies cannot be ascribed to a higher efficiency of vitamin D<sub>2</sub> conversion at exposure during the growth phase. In the present study, mushrooms sampled after overnight cooling and post-harvest exposure were found to have a higher vitamin D<sub>2</sub> content compared to those exposed during the growth phase, and this difference was not caused by effects of overnight cooling on dry matter content (Table 1). The higher conversion of vitamin D<sub>2</sub> in the present study compared with that of Roberts et al. (2008) (6) was probably caused by differences in spectral distribution of the UV-B from different types of lamps. Also, experimental conditions, such as temperature and moisture content, have been shown to affect the formation of vitamin D<sub>2</sub> (5). Orientation of the mushrooms to the UV source has also been a factor that contributes to vitamin D<sub>2</sub> level,



**Fig. 2.** The relationship between vitamin D<sub>2</sub> concentration in mushrooms and exposure to UV-B dose up to 250 mJ cm<sup>-2</sup> in Experiment A and up to 2,400 mJ cm<sup>-2</sup> in Experiment B. Error bars indicate standard errors in Experiment A ( $n=4$ ), and in Experiment B for the highest dose ( $n=6$ ). The equations are linear regression up to 250 and 1,000 mJ cm<sup>-2</sup>. FW = fresh weight.

that is, the content of vitamin D<sub>2</sub> varies within an individual mushroom (8); however, in this study and that of Roberts et al. (2008) (6), the orientations of the mushrooms were similar. Finally, also the difference between our HPLC-DAD and the LC-MS/MS methods used in (6) may give a bias in the results.

The formation of vitamin D<sub>2</sub> levelled off at a content of 164 µg 100 g<sup>-1</sup> at the high dose of 2,400 mJ cm<sup>-2</sup> supplementary UV-B as also reported for doses beyond 1,200 mJ cm<sup>-2</sup> by others (5, 6). In this study, a clear browning effect was observed on the surface of the exposed mushrooms indicating that at high doses of UV-B, there was damage to the surface cells (8). It has been suggested that high doses of UV cause photodegradation of vitamin D<sub>2</sub> (3). Thus, very high doses of UV-B are relatively less efficient for conversion of ergosterol to vitamin D<sub>2</sub> and cause surface discolouration.

UV-B exposure did not affect the growth rate of the mushrooms in Experiment A (Fig. 3). The average yields were 2.25, 4.83 and 3.36 kg m<sup>-2</sup>, and the number of mushrooms were 236, 322 and 163 pieces m<sup>-2</sup> on the first, second and third day of the flush, respectively. The dry matter contents were 8.4, 8.3 and 9.3 g 100 g<sup>-1</sup>, respectively. Over the three days, the vitamin D<sub>2</sub> content of mushrooms decreased sharply from 24 to 3 µg 100 g<sup>-1</sup> when measured after one exposure at 200 mJ cm<sup>-2</sup> of supplementary UV-B (Fig. 4). This decrease was observed mainly from the first to the second day of the experiment. By Day 3, 94% of the vitamin D<sub>2</sub> formed during one exposure to UV-B on Day 1 could be accounted for in the large harvested and the small remaining mushrooms (Table 2). Thus, the fast growth rate of the mushrooms diluted the vitamin D<sub>2</sub> formed after the exposure on Day 1, as the high recovery of vitamin D<sub>2</sub> indicated that there was no significant degradation or transport of vitamin D<sub>2</sub> from the mushrooms.

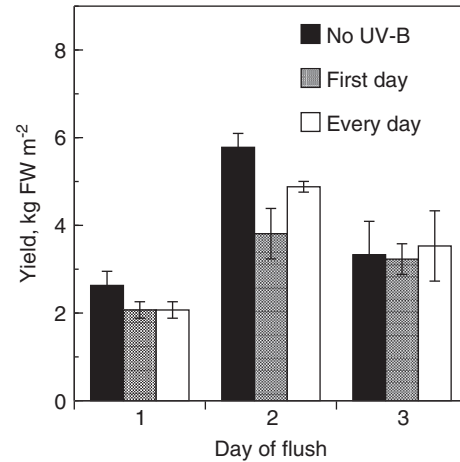
When the UV-B exposure was repeated at daily intervals (Day 1–3), the vitamin D<sub>2</sub> content increased to 33 µg 100 g<sup>-1</sup>. These results suggest that repeated exposure to UV-B is needed just prior to harvest and during all days of a flush to ensure a high level of vitamin D<sub>2</sub> in the final product. The increase in the vitamin D<sub>2</sub> content after repeated exposure may be explained by a carryover of vitamin D<sub>2</sub> formed in the small mushrooms left growing for the next day. There was no indication that one exposure to UV-B stimulated or inhibited the formation of vitamin D<sub>2</sub> at later exposures.

The vitamin D<sub>2</sub> content was much lower in cultivated mushrooms exposed to sunlight of 200 mJ cm<sup>-2</sup> UV-B or in wild mushrooms samples compared to mushrooms exposed to supplementary UV-B (Table 1). The samples of cultivated mushrooms without exposure to UV-B or of miscellaneous origin (strains and producers) were found to have the lowest content of vitamin D<sub>2</sub>. To our knowledge, the vitamin D<sub>2</sub> content of 1.5 µg 100 g<sup>-1</sup> is the first report on wild *Agaricus* sp., while higher contents have been reported for other wild grown species such as *Cantharellus cibarius* and *C. tubaeformis* (10–13 and 21–30 µg 100 g<sup>-1</sup>, respectively) as well as for *Boletus edulis* (3–59 µg 100 g<sup>-1</sup>) (1, 2). The low vitamin D<sub>2</sub> contents of wild *Agaricus* sp. and cultivated *A. bisporus* exposed to 200 mJ cm<sup>-2</sup> UV-B in sunlight were surprising considering the results of other reports and the present study of cultivated *A. bisporus* forming high amounts of vitamin D<sub>2</sub> when exposed to supplementary UV-B (6, 4, 8). Even though *A. bisporus* does not produce as high levels of vitamin D<sub>2</sub> after UV exposure as wild *C. tubaeformis* (2) and cultivated *Pleurotus ostreatus* (5). The difference between the effect of UV-B from sunlight and from the supplementary UV-B unit may have been caused by differences in the distribution of wavelengths within the range of UV-B that are integrated in the measurement of the UV-B sensor. The tubes of the UV-B unit emit some

**Table 1.** The effect of source of UV-B and timing of exposure in relation to harvest on vitamin D<sub>2</sub> in wild and cultivated *Agaricus* mushrooms

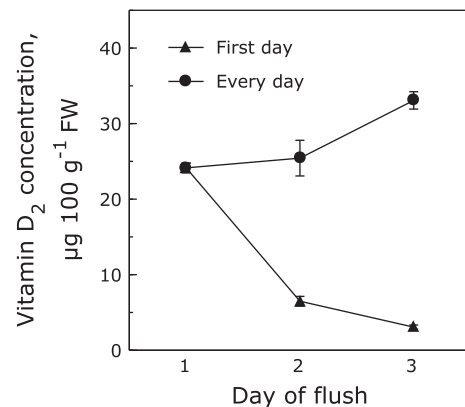
| Origin                  | UV-B source | Relative to harvest | UV-B dose mJ cm <sup>-2</sup> | Dry matter content g 100 g <sup>-1</sup> Fresh weight (FW) | Vitamin D <sub>2</sub> concentration µg 100 g <sup>-1</sup> FW |
|-------------------------|-------------|---------------------|-------------------------------|------------------------------------------------------------|----------------------------------------------------------------|
| Cultivated              | UV-B tubes  | Prior               | 200                           | 8.4                                                        | 24.1 (22.9–25.4) <sup>†</sup> b                                |
| Cultivated              | UV-B tubes  | Post, cooled        | 200                           | 8.6                                                        | 32.4 (27.2–38.7) a                                             |
| Wild                    | Sunlight    | Prior               | 200                           | 12.3                                                       | 1.5 (0.7–2.3) c                                                |
| Cultivated              | Sunlight    | Post                | 200                           | 9.4                                                        | 0.9 (0.8–1.1) c                                                |
| Cultivated              | –           | –                   | 0                             | 10.4                                                       | 0.3 (0.04–0.9) d                                               |
| Cultivated, misc origin | –           | –                   | –                             | 11.5                                                       | 0.2 (0.1–0.4) <sup>‡</sup> d                                   |

<sup>§</sup>Within columns means followed by different letters are significantly different. Means followed by the same letter are not ( $P < 0.05$ ;  $n = 4$ ). <sup>†</sup>Range; <sup>‡</sup>Samples taken from other flush.



**Fig. 3.** The yield of mushrooms obtained during the three days of Experiment A after (a) no exposure to UV-B, exposure to supplementary UV-B of 200 mJ cm<sup>-2</sup>, (b) the first day and (c) repeated every day. Error bars indicate standard errors ( $n = 4$ ). FW = fresh weight.

radiation of wavelengths shorter than 290 nm (Fig. 1), which is the part of the UV-B range that is almost absent in sunlight after absorption in the earth's atmosphere (9). This short-waved UV-B is close to the UV-C range that has, however, been found to be less effective than UV-B in the conversion of ergosterol to vitamin D<sub>2</sub> (5). Reports on effects of sunlight on vitamin D<sub>2</sub> in mushrooms are scarce. *Lentinus edodes* was found to have higher vitamin D<sub>2</sub> content when harvested on sunny as compared to cloudy days (10 as reported in (4)). The production of mushroom usually takes place in dark environment. The UV-exposure is similar to the exposure of wild mushrooms that is not reported to cause toxic elements in mushrooms.



**Fig. 4.** The vitamin D<sub>2</sub> concentration in mushrooms found during the three days of Experiment A after exposure to supplementary UV-B of 200 mJ cm<sup>-2</sup> (a) the first day, and (b) repeated every day. Error bars indicate standard errors ( $n = 4$ ). FW = fresh weight.

**Table 2.** Recovery of vitamin D<sub>2</sub> calculated on basis of production area for three days of experiment A after one exposure to supplementary UV-B of 200 mJ cm<sup>-2</sup> the first day

| Day of experiment | Cumulated harvested mushrooms                        | Remaining mushrooms for next day <sup>a</sup> | Sum                                               |    |
|-------------------|------------------------------------------------------|-----------------------------------------------|---------------------------------------------------|----|
|                   | µg vitamin D <sub>2</sub> m <sup>-2</sup> shelf area |                                               | Recovery of vitamin D <sub>2</sub> <sup>b</sup> % |    |
| 1                 | 499 <sup>c</sup>                                     | 435 <sup>c</sup>                              | 934                                               | —  |
| 2                 | 746                                                  | 98                                            | 844                                               | 90 |
| 3                 | 845                                                  | 31                                            | 876                                               | 94 |

<sup>a</sup>Based on the biomass of small remaining mushrooms and the assumption that their vitamin D<sub>2</sub> content equalled that measured in large harvested mushrooms.

<sup>b</sup>Based on the sum of vitamin D<sub>2</sub> at Day 1 as 100%.

<sup>c</sup>n = 4.

The level of vitamin D<sub>2</sub> content in wild and cultivated mushrooms with or without exposure to sunlight (Table 1) is comparable to that of, for example, 0.4–0.7 µg of vitamin D<sub>3</sub> in 100 g<sup>-1</sup> of pork and beef meat and 0.2–0.4 µg of vitamin D<sub>3</sub> in 100 g<sup>-1</sup> of a variety of cheeses (www.foodcomp.dk).

Commercial growers may consider using UV-B during the growth phase for vitamin D<sub>2</sub> formation in white button mushrooms. In this case, the need for prolonged exposure time is less critical as treatment times do not have to be confined to the time available during the post-harvest processing on a conveyor belt. Instead, an automatic moving UV-B unit can be constructed, for example, for overnight exposure of mushrooms on the growing shelves just prior to harvest. With the high intensities used in the present study, doses of 200 mJ cm<sup>-2</sup> could be obtained with an exposure time of <2 min and resulting in vitamin D<sub>2</sub> contents of around 24 µg 100 g<sup>-1</sup>. At this vitamin D<sub>2</sub> level, approximately 35 g of mushroom is equivalent to the recommended daily intake of 7.5 µg vitamin D<sub>2</sub> for adults in Denmark. The upper safe limit for daily intake of 50 µg vitamin D<sub>2</sub> should be taken into consideration (11). Mushrooms with a vitamin D<sub>2</sub> content of 24 µg 100 g<sup>-1</sup> would contribute to 26% of the recommended daily intake of 7.5 µg based on the average intake of mushrooms by adults of 3 kg yr<sup>-1</sup> in 2006 in Denmark (12). Therefore, UV-B-enhanced vitamin D<sub>2</sub> levels in mushrooms provide a potentially important food source of vitamin D.

## Conclusion

The close to linear relationship between UV-B dose and formation of vitamin D<sub>2</sub> at exposure just prior to harvest makes it possible to produce mushrooms with a well-defined content of vitamin D<sub>2</sub>. Therefore, mushroom products will be a rich source for increased intake of vitamin D<sub>2</sub> in food. Due to the dilution of vitamin D<sub>2</sub> by the fast growth of the mushrooms, UV-B exposure needs to be repeated daily during the harvest flush to maintain

a high content of vitamin D<sub>2</sub> in the end product. Supplementary UV-B from tubes is a more effective source of UV-B for the formation of vitamin D<sub>2</sub> in mushrooms as compared to natural sunlight that gave a low content of vitamin D<sub>2</sub> equivalent to that found in wild *Agaricus* sp. mushrooms.

## Acknowledgements

We thank Astrid Bergmann, Birthe R. Flyger and Kirsten Pinndal for skillful technical assistance. We thank Carsten Dam-Hansen, Technical University of Denmark for providing access to the calibration lamp.

## Conflict of interest and funding

This work was supported by the Danish Food Industry Agency, the Committee for Horticultural Sale and Egehøj Champignon A/S.

## References

- Mattila PH, Piironen VI, Uusi-Rauva EJ, Koivistoinen PE. Vitamin D contents in edible mushrooms. *J Agric Food Chem* 1994; 42: 2449–53.
- Teichmann A, Dutta PC, Staffas A, Jägerstad M. Sterol and vitamin D<sub>2</sub> concentrations in cultivated and wild grown mushrooms: effects of UV irradiation. *LWT* 2007; 40: 815–22.
- Perera CO, Jasinghe VJ. Vitamin D from mushrooms: a review on optimization of the process. *Food Manuf Eff* 2010; 3: 57–67.
- Mau J-L, Chen P-R, Yang J-H. Ultraviolet irradiation increased vitamin D<sub>2</sub> content in edible mushrooms. *J Agric Food Chem* 1998; 46: 5269–72.
- Jasinghe VJ, Perera CO. Ultraviolet irradiation: the generator of vitamin D<sub>2</sub> in edible mushrooms. *Food Chem* 2006; 95: 638–43.
- Roberts JS, Teichert A, McHugh TH. Vitamin D<sub>2</sub> formation from post-harvest UV-B treatment of mushrooms (*Agaricus bisporus*) and retention during storage. *J Agric Food Chem* 2008; 56: 4541–4.
- Jakobsen J, Clausen I, Leth T, Ovesen L. A new method for the determination of vitamin D<sub>3</sub> and 25-OH vitamin D<sub>3</sub> in meat. *J Food Comp Anal* 2004; 17: 777–87.
- Ko JA, Lee BH, Lee JS, Park HJ. Effect of UV-B exposure on the concentration of vitamin D<sub>2</sub> in sliced shiitake mushroom

- (*Lentinus edodes*) and white button mushroom (*Agaricus bisporus*). J Agric Food Chem 2008; 56: 3671–4.
9. Eriksen P. UV radiation and UV-index. Danish Meteorological Institute (in Danish). [http://www.dmi.dk/dmi/uv-straaling\\_og\\_uv-indeks-5.pdf](http://www.dmi.dk/dmi/uv-straaling_og_uv-indeks-5.pdf) [cited 12 November 2010].
  10. Takamura K, Hoshino H. Vitamin D<sub>2</sub> of shiitake mushrooms. N Food Ind 1995; 37: 33–39. (in Japanese)
  11. Nordic Council. Nordic nutrition recommendations, 4th edition. Nordic Council of Ministers, Copenhagen: Nord; 2004, p. 13.
  12. Statistics Denmark. Statbank Denmark. <http://www.dst.dk/HomeUK.aspx> [cited 12 November 2010].

---

\*Hanne L. Kristensen

Department of Food Science  
Aarhus University  
Kirstinebjergvej 10  
DK-5792 Aarslev  
Denmark  
E-mail: Hanne.Kristensen@agrsci.dk